MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, PRECISION TIMERS,
MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, precision timers. Three product assurance classes and a choice of case outline and lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Precision timer
02	Precision timer, dual
03	Precision timer (I_{SOURCE} = -60 mA)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
C	D-1 (14-lead, 1/4" x 3/4", dual-in-line package)
G	A-1 (8-lead can)
P	D-4 (8-lead, 1/4" x 3/8", dual-in-line package)

1.3 Absolute maximum ratings.

Positive supply voltage - - - - - - - - - - - - - - - - +18 Vdc
Discharge current - - - - - - - - - - - - - - - - - - - +200 mA
Output sink current - - - - - - - - - - - - - - - - - - +200 mA
Output source current:
  Device types 01 and 02 - - - - - - - - - - - - - - - -200 mA
  Device type 03 -60 mA
Storage temperature range - - - - - - - - - - - - - - -65°C to +150°C
Lead temperature (soldering, 60 seconds) - - - - - - - 300°C
Junction temperature - - - - - - - - - - - - - - - - - T_J = 175°C

1.4 Recommended operating conditions.

Supply voltage range:
  Device types 01 and 03 - - - - - - - - - - - - - - - +4.5 to +16.5 Vdc
  Device type 02 - - - - - - - - - - - - - - - - - - - +5.0 to +15.0 Vdc 1/
Case operating temperature range - - - - - - - - - - - -55°C to +125°C

1/ Supply voltage upper limit due to excessive junction temperature at T_C = 125°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.5 Power and thermal characteristics.

Case outline	Package	Maximum allowable power dissipation	Maximum θ J-C	Maximum θ J-A
C	14-lead dual-in-line	400 mW at T_C = 125°C	45°C/W	120°C/W
G	8-lead can	300 mW at T_C = 125°C	40°C/W	150°C/W
P	8-lead dual-in-line	370 mW at T_C = 125°C	45°C/W	135°C/W

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Schematic circuits. The schematic circuits shall be as specified on figure 2.

3.2.3 Block diagram and circuit operation table. The block diagram and circuit operation table shall be as specified on figure 3.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The following electrical performance characteristics apply over the full operating case temperature range of -55° to +125°C and for supply voltages of +4.5 Vdc to +16.5 Vdc for device types 01 and 03, and +5.0 Vdc to +15.0 Vdc for device type 02, unless otherwise specified (see table I). See figure 3 for circuit operation table.

3.4.1 Triggering. In the monostable mode, the device is triggered on the negative slope of a trigger pulse. The trigger pulse must be of shorter duration than the "RC" time interval. The minimum trigger pulse width is 1 microsecond (s).

Text continues on page 10.

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TABLE I. Electrical performance characteristics.
$-55°C \leq T_C \leq 125°C$ unless otherwise specified

Characteristic	Symbol	Conditions (Figure 6 unless otherwise specified)			Limits 1/ Min	Limits 1/ Max	Limits 1/ Units
Power supply current	I_{CC}	$R_L = \infty$	V_{CC} = 4.5 Vdc for device types 01, 03		-	5.0	mA
			V_{CC} = 5.0 Vdc for device type 02		-	10.0	mA
		$R_L = \infty$	V_{CC} = 16.5 Vdc for device types 01, 03		-	20.0	mA
			V_{CC} = 15.0 Vdc for device type 02		-	30.0	mA
Trigger voltage	V_{TR}	V_{CC} = 4.5 Vdc for device types 01,03 V_{CC} = 5.0 Vdc for device type 02	T_C = 25°C	for device types 01, 03	1.30	1.80	V
				for device type 02	1.45	1.95	V
			T_C = -55°C	for device types 01, 03	1.15	1.80	V
				for device type 02	1.30	1.95	V
			T_C = 125°C	for device types 01, 03	1.30	2.10	V
				for device type 02	1.45	2.25	V
		V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15.0 Vdc for device type 02	T_C = 25°C	for device types 01, 03	5.20	5.80	V
				for device type 02	4.70	5.30	V
			T_C = -55°C	for device types 01, 03	5.00	5.80	V
				for device type 02	4.50	5.30	V
			T_C = 125°C	for device types 01, 03	5.20	6.10	V
				for device type 02	4.70	5.60	V
Trigger current	I_{TR}	V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15.0 Vdc for device type 02 V_{TR} = 0 Vdc			-5.0	-	µA
Threshold voltage	V_{TH}	V_{CC} = 4.5 Vdc for device types 01,03 V_{CC} = 5.0 Vdc for device type 02	T_C = 25°C	for device types 01, 03	2.70	3.30	V
				for device type 02	3.05	3.65	V
			$-55°C \leq T_C \leq 125°C$	for device types 01, 03	2.60	3.40	V
				for device type 02	2.95	3.75	V
		V_{CC} = 16.5 Vdc for device types 01, 03 V_{CC} = 15.0 Vdc for device type 02	T_C = 25°C	for device types 01, 03	10.70	11.30	V
				for device type 02	9.70	10.30	V
			$-55°C \leq T_C \leq 125°C$	for device types 01, 03	10.60	11.40	V
				for device type 02	9.60	10.40	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics. - Continued
$-55°C \leq T_C \leq 125°C$ unless otherwise specified

Characteristic	Symbol	Conditions (Figure 6 unless otherwise specified)		Limits 1/ Min	Limits 1/ Max	Limits 1/ Units
Threshold current	I_{TH}	V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15.0 Vdc for device type 02	$25° \leq T_C = 125°C$	-	250	nA
			$T_C = -55°C$	-	2.5	µA
Low level output voltage	V_{OL}	V_{CC} = 4.5 Vdc for device types 01,03 V_{CC} = 5.0 Vdc for device type 02 I_{sink} = 5 mA	$T_C = 25°C$	-	.250	V
			$T_C = -55°C$, $T_C = 125°C$	-	.350	V
		V_{CC} = 4.5 Vdc for device types 01,03 V_{CC} = 5.0 Vdc for device type 02 I_{sink} = 50 mA	$T_C = -55°C$	-	2.60	V
			$25°C \leq T_C \leq 125°C$	-	2.20	V
		V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15 Vdc for device type 02 I_{sink} = 10 mA	$-55°C \leq T_C \leq 25°C$	-	.150	V
			$T_C = 125°C$	-	.250	V
		V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15 Vdc for device type 02 I_{sink} = 50 mA	$-55°C \leq T_C \leq 25°C$	-	.500	V
			$T_C = 125°C$	-	.700	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics. - Continued
$-55°C \leq T_C \leq 125°C$ unless otherwise specified

Characteristic	Symbol	Conditions (Figure 6 unless otherwise specified)			Limits 1/ Min	Limits 1/ Max	Limits 1/ Units
Low level output voltage	V_{OL}	V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15.0 Vdc for device type 02 I_{sink} = 100 mA	T_C = 25°C		-	2.20	V
			T_C = -55°C, T_C = 125°C		-	2.80	V
High level output voltage	V_{OH}	V_{CC} = 4.5 Vdc for device types 01,03 V_{CC} = 5.0 Vdc for device type 02 I_{source} = -100 mA 6/	25°C ≤ T_C ≤ 125°C	for device types 01, 03	2.60	-	V
				for device type 02	3.10	-	V
			T_C = -55°C	for device types 01, 03	2.20	-	V
				for device type 02	2.70	-	V
		V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15.0 Vdc for device type 02 I_{source} = -100 mA 6/	25°C ≤ T_C ≤ 125°C	for device types 01, 03	14.6	-	V
				for device type 02	13.1	-	V
			T_C = -55°C	for device types 01, 03	14.0	-	V
				for device type 02	12.5	-	V
Discharge transistor leakage current	I_{CEX}	V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15.0 Vdc for device type 02	-55°C ≤ T_C ≤ 25°C		-	100	nA
			T_C = 125°C		-	3.0	µA
Discharge transistor saturation voltage	V_{SAT}	V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15.0 Vdc for device type 02 I_D = 50 mA	-55°C ≤ T_C ≤ 25°C		-	0.80	V
			T_C = 125°C		-	1.00	V
Reset voltage	V_R	V_{CC} = 16.5 Vdc for device types 01,03 V_{CC} = 15.0 Vdc for device type 02 (see Figure 10)			0.1	1.3	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics. - Continued
$-55°C \leq T_C \leq 125°C$ unless otherwise specified

Characteristic	Symbol	Conditions (Figure 6 unless otherwise specified)		Limits 1/ Min	Limits 1/ Max	Limits 1/ Units
Reset current	I_R	V_{CC} = 16.5 Vdc for device types 01, 03 V_{CC} = 15.0 Vdc for device type 02 V_R = 0 Vdc		-1.6	0	mA
Propagation delay time, low to high level output (monostable)	t_{PLH}	4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device type 02 R_T = 1 kΩ C_T = 0.1 µF (see Figure 7)	$-55°C \leq T_C \leq 25°C$	-	800	ns
			T_C = 125°C	-	900	ns
Transition time, low to high level output (monostable)	t_{TLH}	4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device type 02 R_T = 1 kΩ C_T = 0.1 µF (see Figure 7)		-	300	ns
Transition time, high to low level output (monostable)	t_{THL}	4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device 02 R_T = 1 kΩ C_T = 0.1 µF (see Figure 7)		-	300	ns
Time delay, output high (monostable)	$t_{D(OH)}$	4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device type 02 R_T = 1 kΩ C_T = 0.1 µF (see Figure 7)		106.7	113.3	µs
Time delay, output high (monostable)	$t_{D(OH)}$	4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device type 02 R_T = 100 kΩ C_T = 0.1 µF (see Figure 7)		10.67	11.33	ms

See footnotes at end of table.

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TABLE I. Electrical performance characteristics. - Continued
$-55°C \leq T_C \leq 125°C$ unless otherwise specified

Characteristic	Symbol	Conditions (Figure 6 unless otherwise specified)	Limits 1/ Min	Limits 1/ Max	Limits 1/ Units
Drift in time delay vs change in supply voltage (monostable)	$\frac{\Delta t_{D(OH)}}{\Delta V_{CC}}$	ΔV_{CC} = 12 Vdc for device types 01, 03 ΔV_{CC} = 10 Vdc for device type 02 R_T = 1 kΩ C_T = 0.1 µF T_C = 25°C (see Figure 7)	-220	220	ns/V
Propagation delay time, threshold to output	t_{PHL}	4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device type 02 R_T = 1 kΩ (see Figure 8)	-	12.0	µs
Temperature coefficient of time delay (monostable)	$\frac{\Delta t_{D(OH)}}{\Delta T}$	V_{CC} = 16.5 Vdc for device types 01, 03 V_{CC} = 15.0 Vdc for device type 02 R_T = 1 kΩ C_T = 0.1 µF (see Figure 7)	-11	11	ns/°C
Capacitor charge time (astable)	t_{ch}	4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device type 02 $R_{TA} = R_{TB}$ = 1 kΩ C_T = 0.1 µF (see Figure 9)	120	156	µs
		4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device type 02 $R_{TA} = R_{TB}$ = 100 kΩ C_T = 0.1 µF (see Figure 9)	11.3	15.0	ms
Capacitor discharge time (astable)	t_{dis}	4.5 Vdc $\leq V_{CC} \leq$ 16.5 Vdc for device types 01, 03 5.0 Vdc $\leq V_{CC} \leq$ 15.0 Vdc for device type 02 $R_{TA} = R_{TB}$ = 1 kΩ C_T = 0.1 µF (see Figure 9)	57.5	80	µs

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TABLE I. Electrical performance characteristics. - Continued
$-55°C \leq T_C \leq 125°C$ unless otherwise specified

Characteristic	Symbol	Conditions (Figure 6 unless otherwise specified)		Limits 1/ Min	Max	Units
Capacitor discharge time (astable)	t_{dis}	$4.5\ Vdc \leq V_{CC} \leq 16.5\ Vdc$ for device types 01, 03 $5.0\ Vdc \leq V_{CC} \leq 15.0\ Vdc$ for device type 02 $R_{TA} = R_{TB} = 100\ k\Omega$ $C_T = 0.1\ \mu F$ (see Figure 9)		5.4	7.7	ms
Drift in capacitor charge time vs change in supply voltage (astable)	$\frac{\Delta t_{ch}}{\Delta V_{CC}}$	$\Delta V_{CC} = 12$ Vdc for device types 01, 03 $\Delta V_{CC} = 10$ Vdc for device type 02 $R_{TA} = R_{TB} = 1\ k\Omega$ $C_T = 0.1\ \mu F$ (see Figure 9)	$T_C = 25°C$	-820	820	ns/V
Temperature coefficient of capacitor charge time (astable)	$\frac{t_{ch}}{\Delta T}$	$V_{CC} = 16.5$ Vdc for device types 01, 03 $V_{CC} = 15.0$ Vdc for device type 02 $R_{TA} = R_{TB} = 1\ k\Omega$ $C_T = 0.1\ \mu F$ (see Figure 9)		-68	68	ns/°C
Reset time	t_{res}	$V_{CC} = 16.5$ Vdc for device types 01, 03 $V_{CC} = 15.0$ Vdc for device type 02 (see Figure 10)	$-55°C \leq T_C \leq 25°C$	-	1.5	µs
			$T_C = 125°C$	-	2	µs
Matching, time delay, output high (monostable)	$\Delta t_{D(OH)M}$ 3/	$5\ Vdc \leq V_{CC} \leq 15\ Vdc$ $R_T = 1\ k\Omega$ $C_T = 0.1\ \mu F$ $T_C = 25°C$ 2/ (see Figure 7)		-1.1	+1.1	µs
Matching, temperature coefficient of time delay (monostable)	$\frac{\Delta t_{D(OH)M}}{\Delta T}$ 4/	$V_{CC} = 15$ Vdc $R_T = 1\ k\Omega$ $C_T = 0.1\ \mu F$ 2/ (see Figure 7)		-20	+20	ns/°C

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TABLE I. Electrical performance characteristics. - Continued
$-55°C \leq T_C \leq 125°C$ unless otherwise specified

Characteristic	Symbol	Conditions (Figure 6 unless otherwise specified)	Limits 1/		
			Min	Max	Units
Matching, drift in time delay vs supply voltage (monostable)	$\frac{\Delta t_{D(OH)M}}{\Delta V_{CC}}$ 5/	ΔV_{CC} = 10 V R_T = 1 kΩ C_T = 0.1 µF T_C = 25°C 2/ (see Figure 7)	-110	+110	ns/V

1/ Limits apply to both device types unless otherwise specified. Each side of device type 02 shall be tested separately.

2/ Device type 02 only - matching between side "A" and side "B".

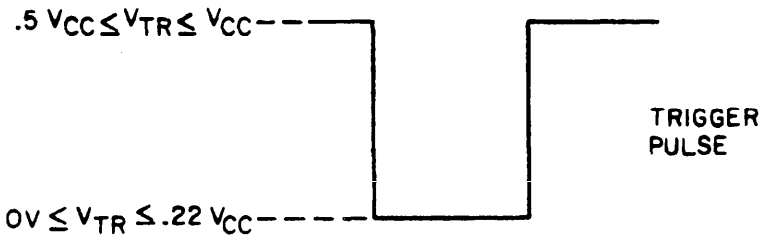
3/ $\Delta t_{D(OH)M} = t_{D(OH)}$ side "A" - $t_{D(OH)}$ side "B".

4/ $\frac{\Delta t_{D(OH)M}}{\Delta T} = \frac{\Delta t_{D(OH)}}{\Delta T}$ side "A" - $\frac{\Delta t_{D(OH)}}{\Delta T}$ side "B".

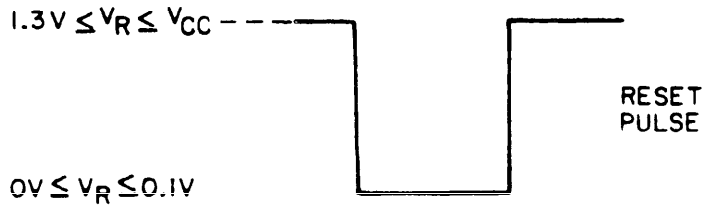
5/ $\frac{\Delta t_{D(OH)M}}{\Delta V_{CC}} = \frac{\Delta t_{D(OH)}}{\Delta V_{CC}}$ side "A" - $\frac{\Delta t_{D(OH)}}{\Delta V_{CC}}$ side "B".

6/ I_{source} for device type 03 = -60 mA.

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3.4.2 Reset. In the monostable mode, the device may be reset (from V_{OH} to V_{OL}) on the negative slope of a reset pulse. Once the reset is returned high, the output will remain low only if the trigger is high. If the trigger is low when the reset is returned high, the output will go high. The minimum reset pulse width is 2 µs.



3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3,9	1*,2,3,9 1/	1
Group A test requirements (method 5005)	1,2,3,9, 10,11	1,2,3,9	1,2,3,9
Group C end-point and group B, class S, electrical parameters (method 5005)	1,2,3, and table IV delta limits	1 and table IV delta limits	Table IV delta limits
Additional electrical subgroups for group C periodic inspections	Not applicable	10,11	None
Group D end-point electrical parameters (method 5005)	1,2,3	1	None

* PDA applies to subgroup 1 (see 4.2d).

1/ Test numbers 58 through 65, and 83 are not required.

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3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 54 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883):
   1. For class S devices, test condition C using the circuit shown on figure 4.
   2. For class B devices, test condition C using the circuit shown on figure 4, or test condition F using the circuit shown on figure 5.
b. Reverse bias burn-in (method 1015 of MIL-STD-883) - For class S devices only, using the circuit shown on figure 4.
c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
d. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Subgroups 4, 5, 6, 7, and 8 shall be omitted.
b. Tests shall be as specified in table II herein.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

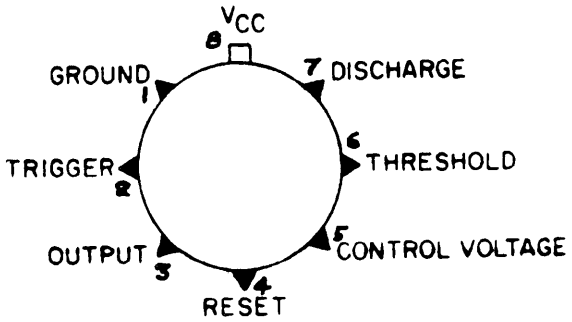
a. End-point electrical parameters shall be as specified in table II herein.
b. Life test for class S devices shall be in accordance with table IIa (subgroup 5) of method 5005 of MIL-STD-883, using the circuit shown on figure 5. If the alternate burn-in conditions are used, the circuits on figure 4 shall be used.

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Device types 01 and 03

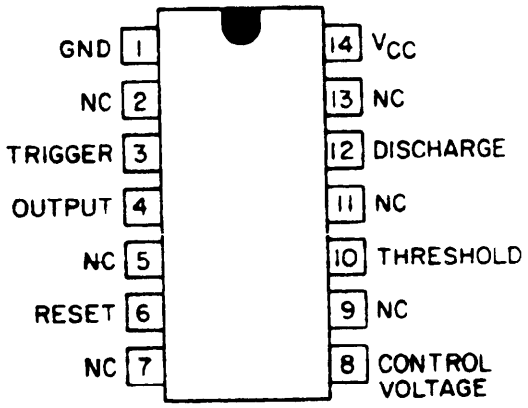
Case G



8 LEAD CAN (TOP VIEW)

Device types 01 and 03

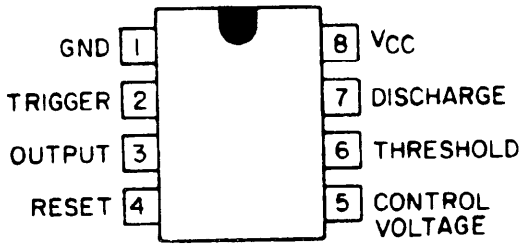
Case C



14 LEAD DUAL-IN-LINE
(TOP VIEW)

Device types 01 and 03

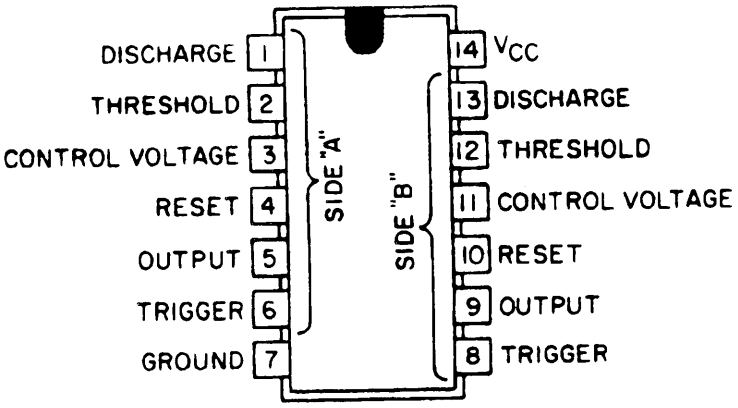
Case P



8 LEAD DUAL-IN-LINE
(TOP VIEW)

Device type 02 SEE NOTE

Case C



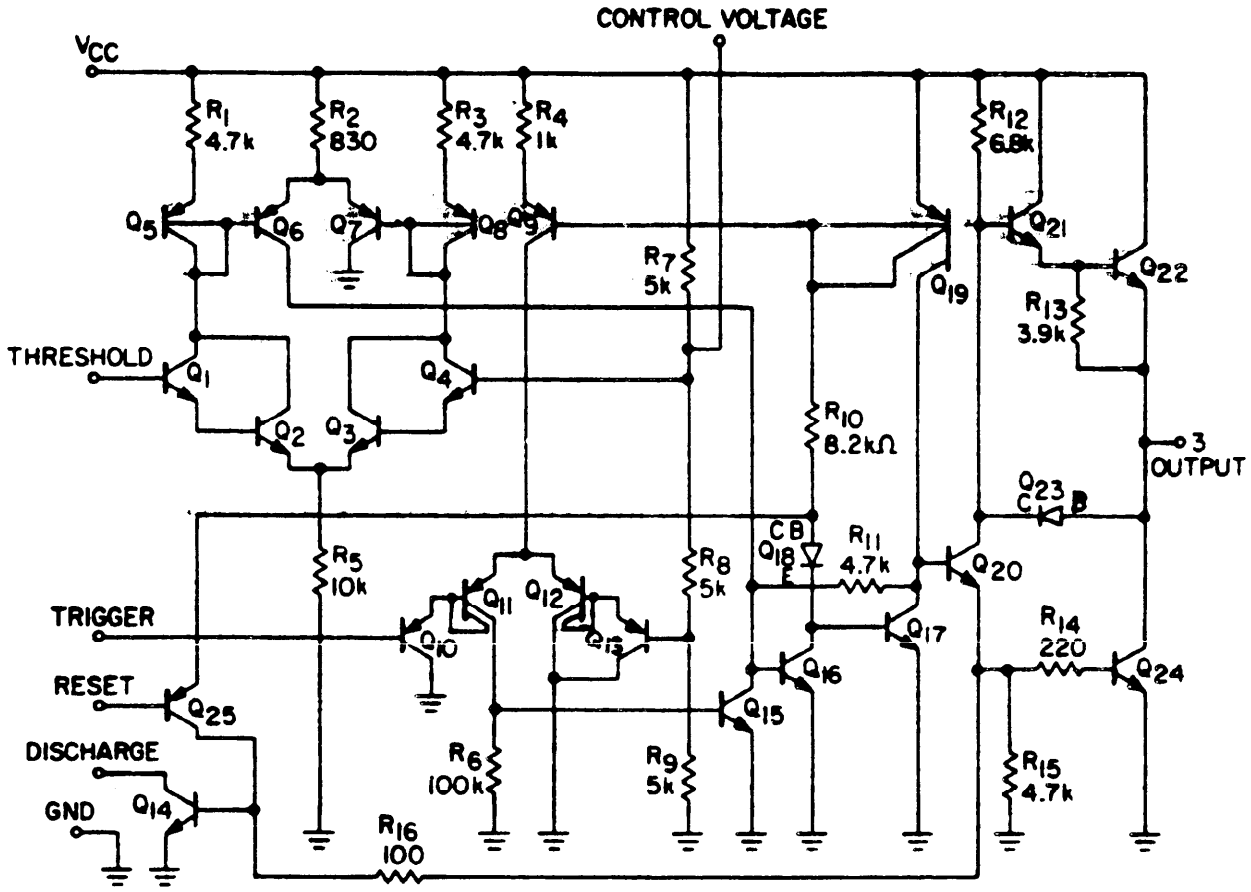
14 LEAD DUAL-IN-LINE
(TOP VIEW)

NOTE: V_{CC} and GROUND are common to both sides.

FIGURE 1. Terminal connections.

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Device types 01 and 03

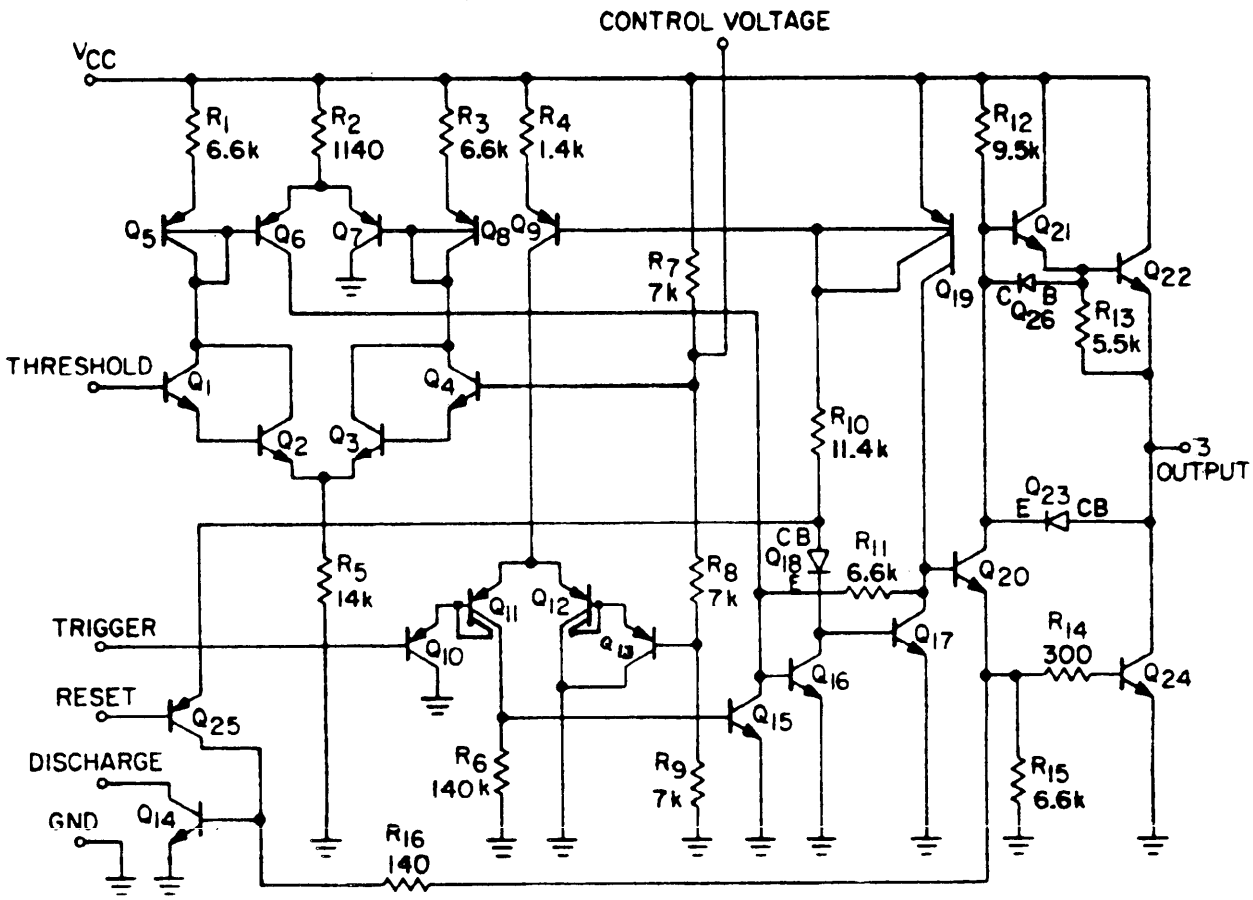


Circuit A

FIGURE 2. Schematic circuit.

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Device type 02



Circuit A

NOTE: For device type 02, two circuits are in the package.

FIGURE 2. Schematic circuit - Continued.

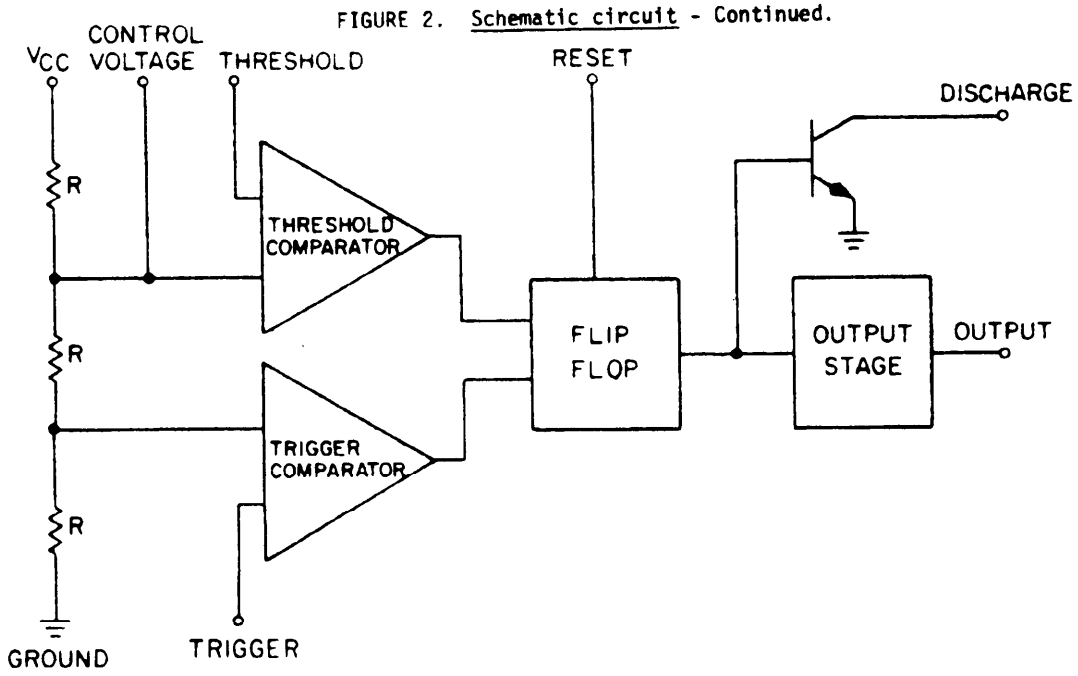


FIGURE 3. Block diagram and circuit operation table.

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Check the source to verify that this is the current version before use.

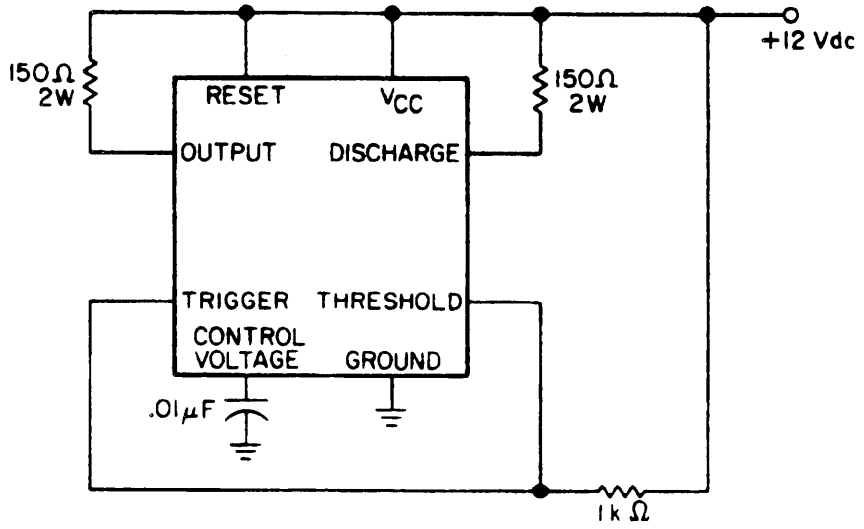
INPUTS			OUTPUT
$\overline{\text{RESET}}$	THRESHOLD	$\overline{\text{TRIGGER}}$	
⊓	0	1	RESETS (⊓)
⊓	1	1	0
⊓	0	0	⊓
⊓	1	0	⊓
1	⊓	1	RESETS
1	⊓	0	1
0	⊓	1	0
0	⊓	0	0
1	0	⊓	SETS (⊔)
1	1	⊓	⊔ (SEE NOTE 1)
0	0	⊓	0
0	1	⊓	0

NOTES:
1. Some devices latch high for V_{CC} < 10 Vdc.
2. Discharge transistor follows the output as follows:
   Output high = Discharge transistor OFF
   Output low = Discharge transistor ON

FIGURE 3. Block diagram and circuit operation table - Continued.

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Device types 01 and 03



Device type 02

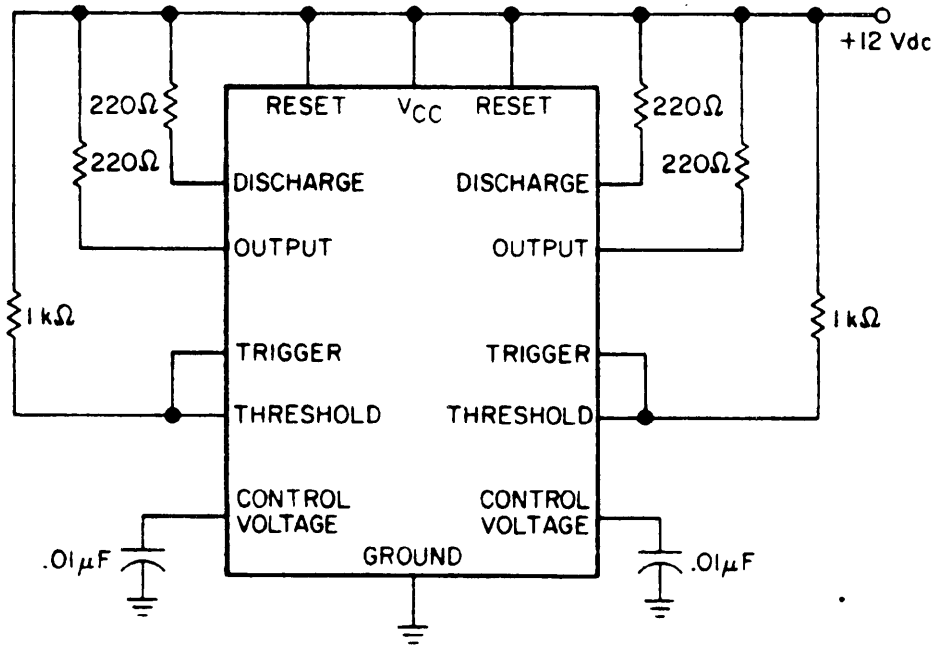
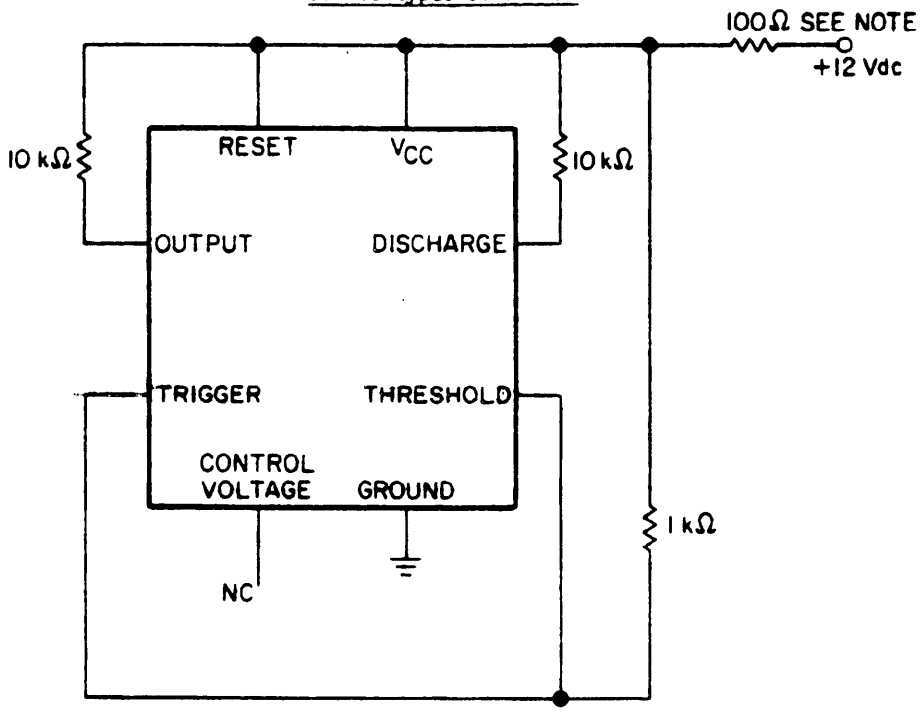


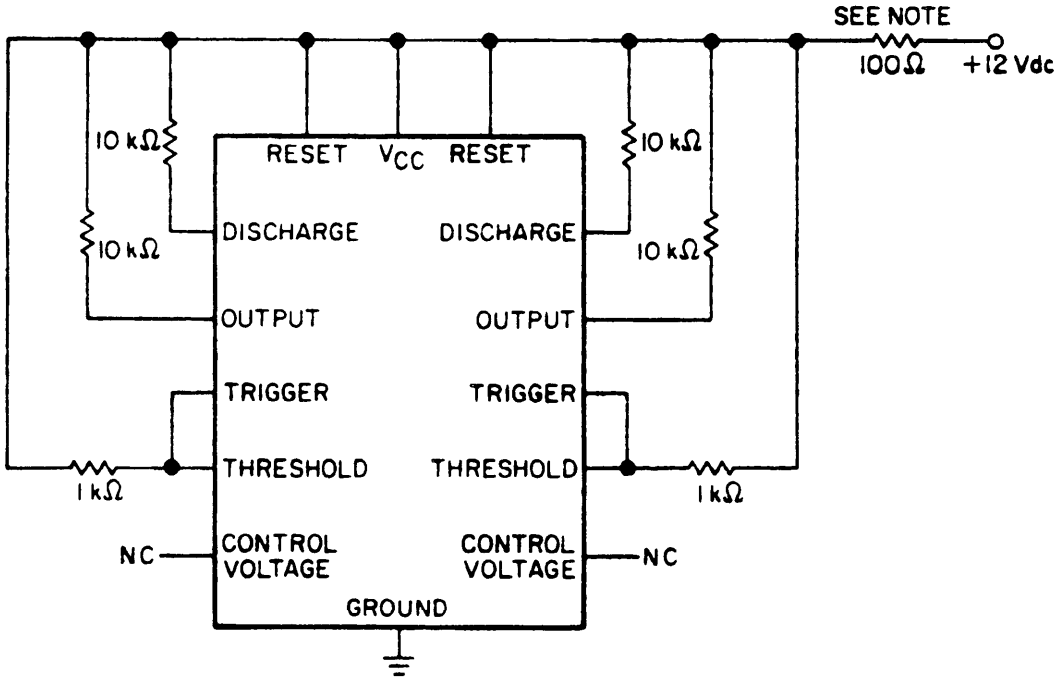
FIGURE 4. Burn-in and operating life test circuit.

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Device types 01 and 03



Device type 02

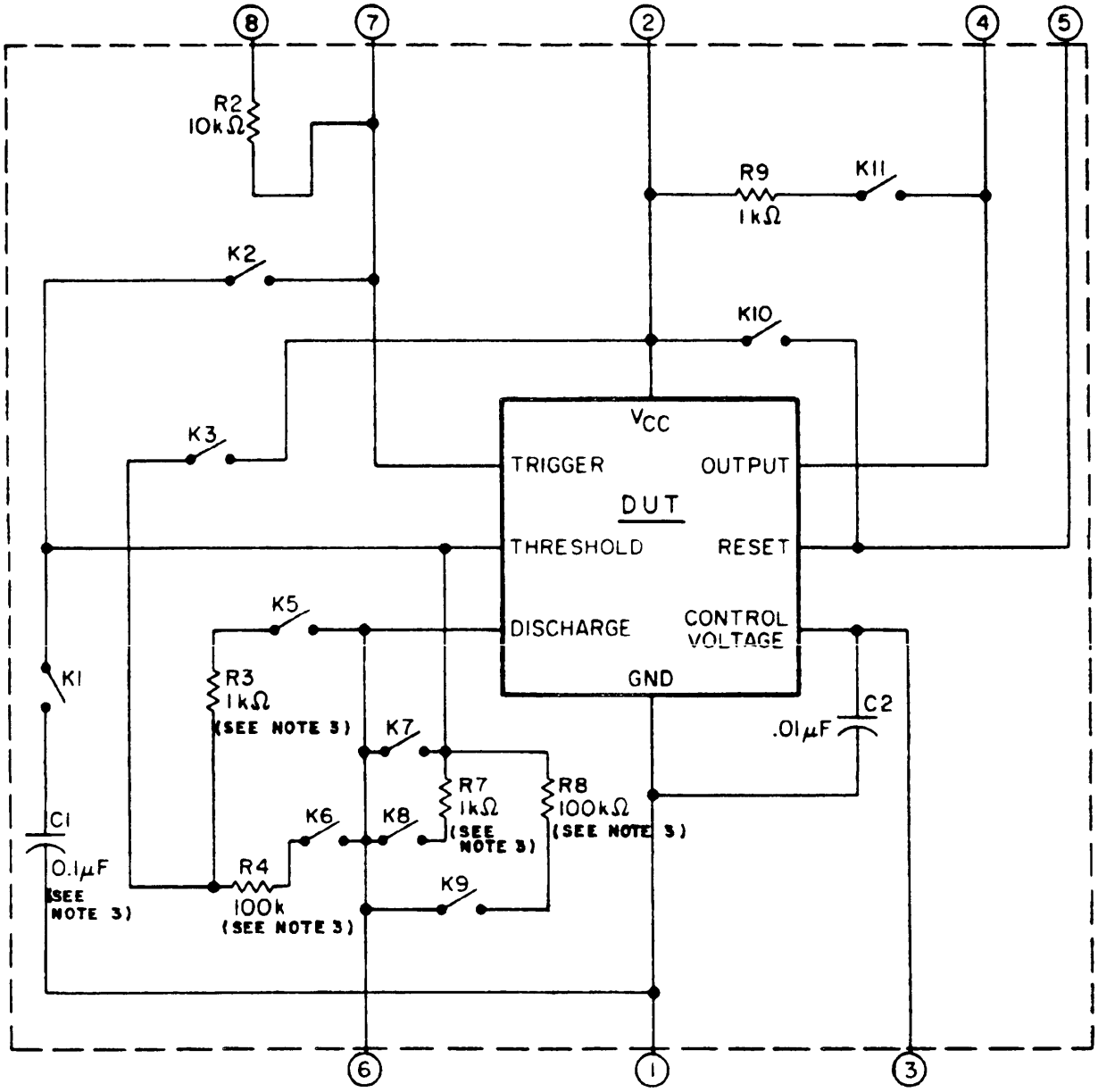


NOTE: The 100Ω resistor is optional. If accelerated high temperature test conditions are used, the device manufacturer shall ensure that at least 85% of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

FIGURE 5. Accelerated burn-in and operating life test circuit.

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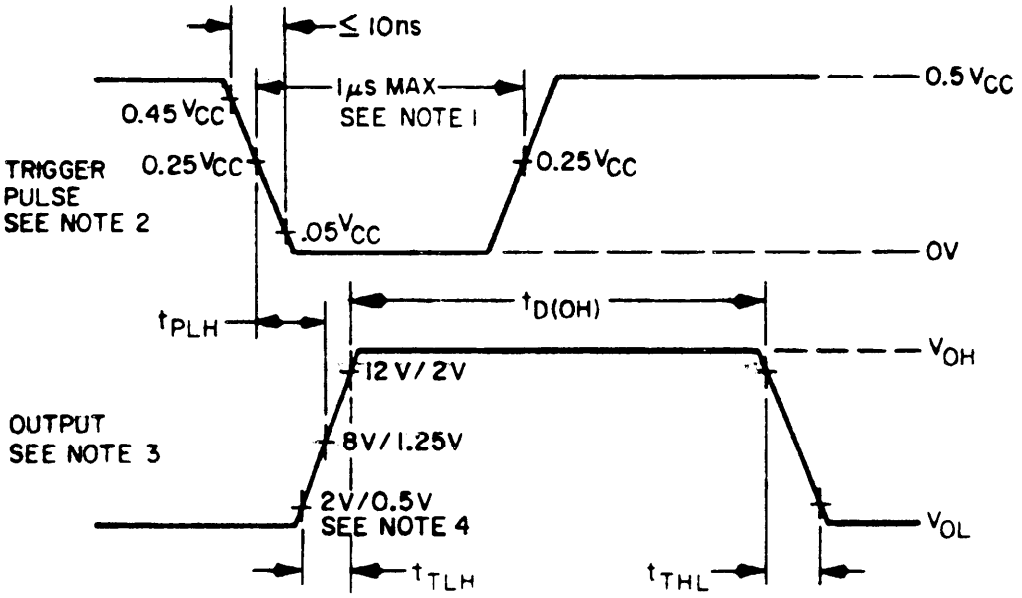
TEST CIRCUIT PIN OUT
SEE NOTE 1



NOTES:
1. Test circuit pin conditions and test temperatures shall be as specified in table III.
2. Precautions shall be taken to prevent damage to the D.U.T. during insertion into the socket and during changing of relay switching positions (e.g. disable voltage/current supplies, current limit, $\pm V_{CC}$, etc.).
3. These resistors and capacitor are 0.1% tolerance. The remainder of resistors and capacitors are 10% tolerance.

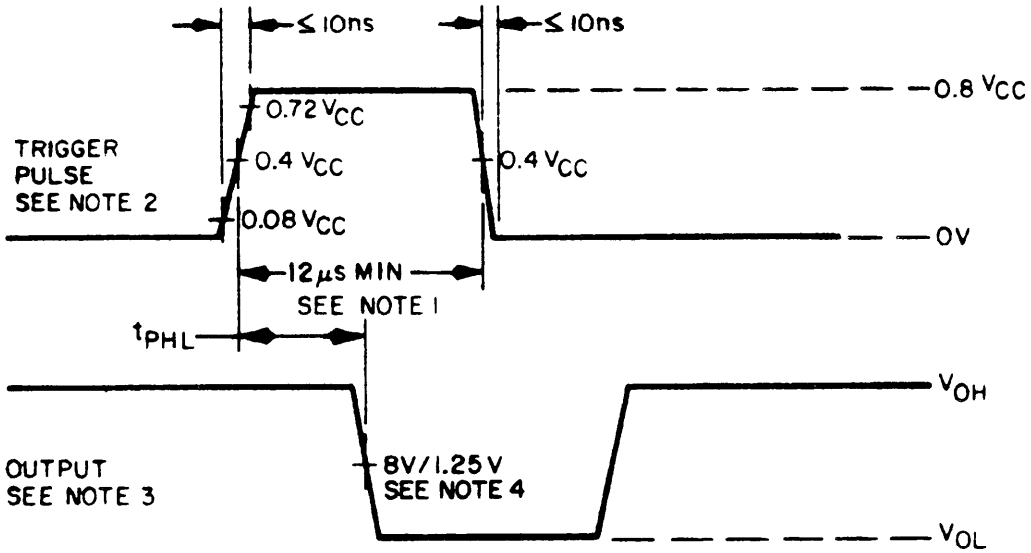
FIGURE 6. Test circuit pin out.

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NOTES:
1. This limit applies only during tests for t_{PLH}, t_{TLH} and t_{THL}.
2. This pulse is applied to pin 7 of the test adapter.
3. This measurement shall be made on pin 4 of the test adapter.
4. A/B means value "A" for V_{CC} = 15 V or 16.5 V and "B" for V_{CC} = 4.5 V or 5 V.

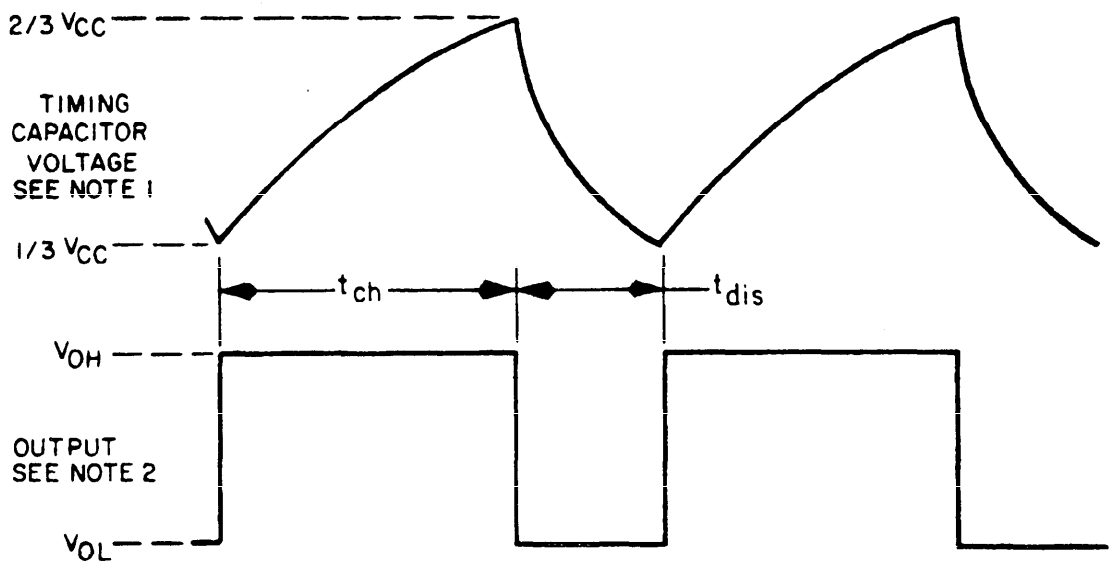
FIGURE 7. Waveforms for switching and timing parameters (monostable).



NOTES:
1. This limit applies only during tests for t_{PHL}.
2. This pulse is applied to pin 7 of the test adapter.
3. This measurement shall be made on pin 4 of the test adapter.
4. A/B means value "A" for V_{CC} = 15 V or 16.5 V and "B" for V_{CC} = 4.5 V or 5 V.

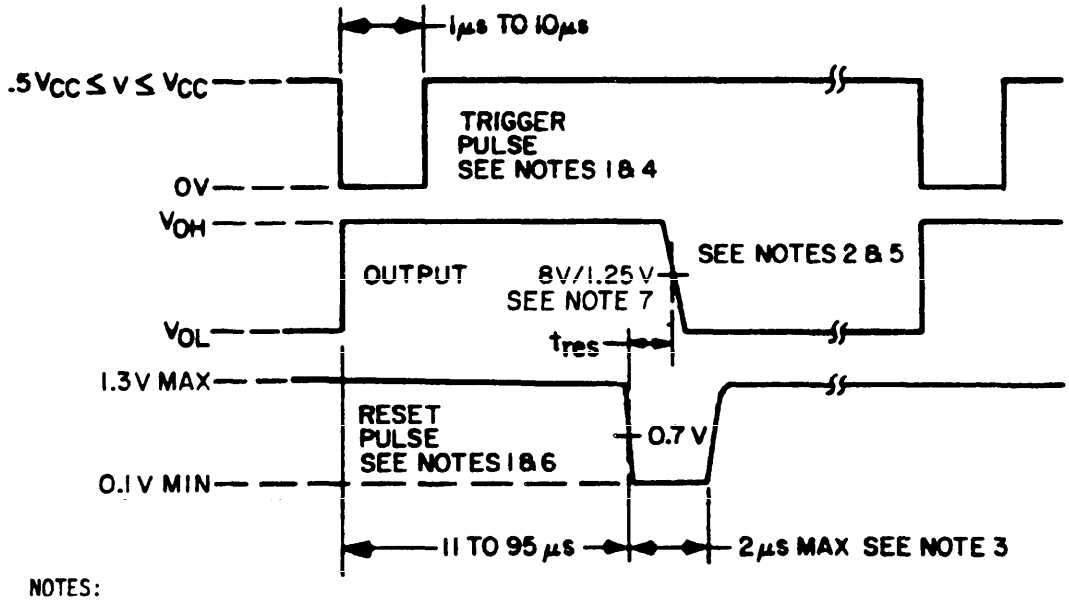
FIGURE 8. Waveforms for switching and timing parameters (monostable).

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Check the source to verify that this is the current version before use.



NOTES:
1. The timing capacitor voltage measurement (threshold) shall be made on pin 7 of the test adapter.
2. The timing measurement (output) shall be made at pin 4 of the test adapter.

FIGURE 9. Waveforms for timing parameters (astable).



NOTES:
1. Reset pulse rise and fall times shall be ≤ 10 ns.
2. Reset thresholds are checked by applying the specified reset levels and verifying

   a. Output high after trigger pulse and before reset goes low and
   b. Output goes low and stays low 2 µsec min. after reset is returned high.

3. This limit only applies during test for t_{RES}.
4. This pulse shall be applied to pin 7 of the test adapter.
5. This measurement shall be made on pin 4 of the test adapter.
6. This pulse shall be applied to pin 5 of the test adapter.
7. A/B means value "A" for V_{cc} = 15 V or 16.5 V and "B" for V_{cc} = 4.5 V or 5 V.

FIGURE 10. Waveforms for reset tests.

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TABLE III. Group A inspection for device types 01, 02, and 03.

Subgroup	Symbol	MIL-STD-883 method	Test No.	Adapter pin numbers 14/ 1	2 Device 01,03	2 Device 02	3	4	5	6 Device 01,03	6 Device 02	7	8 Device 01,03	8 Device 02	Relays closed 14/	Pin measured No.	Pin measured Value	Pin measured Unit	Device 01, 03 limits Min	Device 01, 03 limits Max	Device 01, 03 limits Unit	Device 02 limits 1/ Min	Device 02 limits 1/ Max	Device 02 limits 1/ Unit
					Volts, dc								Volts, dc											
1 T_C = 25°C	I_{CC}	3005	1	GND	4.5	5							3.5	3.8	2, 10	2	I_{CC}	mA	---	5.0	mA	---	10.0	mA
		3005	2		16.5	15							11.5	10.5	2, 10	2	I_{CC}	mA	---	20.0	mA	---	30.0	mA
	V_{TR}		3		4.5	5		2/					2/	2/	2,3,5,10	11	V_{TR}	V	1.30	1.80	V	1.45	1.95	V
			4		16.5	15		2/					2/	2/			V_{TR}	V	5.20	5.80	V	4.70	5.30	V
	I_{TR}		5		16.5	15							5.0	4.5			I_{TR}	µA	-5.0	---	µA	-5.0	---	µA
	V_{TH}		6		4.5	5		3/					3/	3/			V_{TH}	V	2.70	3.30	V	3.05	3.65	V
			7		16.5	15		3/					3/	3/			V_{TH}	V	10.70	11.30	V	9.70	10.30	V
	I_{TH}		8		16.5	15							11.5	10.5			I_{TH}	nA	---	250	nA	---	250	nA
	V_{OL}	3007	9		4.5	5		5 mA					3.5	3.8		4 15/	V_O	V	---	0.250	V	---	0.250	V
			10		4.5	5		50 mA					3.5	3.8					---	2.20		---	2.20	
			11		16.5	15		10 mA					11.5	10.5					---	0.150		---	0.150	
			12					50 mA											---	0.500		---	0.500	
			13					100 mA											---	2.20		---	2.20	
	V_{OH}	3006	14		4.5	5		-100 mA 16/					1.0	1.15					2.60	---		3.10	---	
		3006	15		16.5	15		-100 mA 16/					5.0	4.5					14.6	---		13.1	---	
	I_{CEX}		16		16.5	15				16.5 V	15.0 V		5.0	4.5	2,10	6	I_O	nA	---	100	nA	---	100	nA
	V_{SAT}		17		16.5	15				50 mA	50 mA		11.5	10.5	2,10	6	V_{SAT}	V	---	0.80	V	---	0.80	V
	V_R		18		16.5	15		See fig. 10	See fig. 10			See fig. 10			1,3,5,7,11	5	V_R	V	0.10	1.30	V	0.10	1.30	V
	I_R		19		16.5	15			4/				5.0	4.5	2,3,5	5	I_R	mA	-1.60	---	mA	-1.60	---	mA
2 T_C = 125°C	I_{CC}	3005	20		4.5	5							3.5	3.8	2,10	2	I_{CC}	mA	---	5.0	mA	---	10.0	mA
		3005	21		16.5	15							11.5	10.5	2,10	2	I_{CC}	mA	---	20.0	mA	---	30.0	mA
	V_{TR}		22		4.5	5		2/					2/	2/	2,3,5,10	11	V_{TR}	V	1.30	2.10	V	1.45	2.25	V
			23		16.5	15		2/					2/	2/			V_{TR}	V	5.20	6.10	V	4.70	5.60	V
	I_{TR}		24		16.5	15							5.0	4.5			I_{TR}	µA	-5.0	---	µA	-5.0	---	µA
	V_{TH}		25		4.5	5		3/					3/	3/			V_{TH}	V	2.60	3.40	V	2.95	3.75	V
			26		16.5	15		3/					3/	3/			V_{TH}	V	10.60	11.40	V	9.60	10.40	V
	I_{TH}		27		16.5	15							11.5	10.5			I_{TH}	nA	---	250	nA	---	250	nA
	V_{OL}	3007	28		4.5	5		5 mA					3.5	3.8		4 15/	V_O	V	---	0.350	V	---	0.350	V
			29		4.5	5		50 mA					3.5	3.8					---	2.20		---	2.20	
			30		16.5	15		10 mA					11.5	10.5					---	0.250		---	0.250	
			31					50 mA											---	0.700		---	0.700	
			32					100 mA											---	2.80		---	2.80	
	V_{OH}	3006	33		4.5	5		-100 mA 16/					1.0	1.15					2.60	---		3.10	---	
		3006	34		16.5	15		-100 mA 16/					5.0	4.5					14.6	---		13.1	---	
	I_{CEX}		35							16.5 V	15.0 V		5.0	4.5	2,10	6	I_O	µA	---	3.0	µA	---	3.0	µA
	V_{SAT}		36							50 mA	50 mA		11.5	10.5	2,10	6	V_{SAT}	V	---	1.00	V	---	1.00	V
	V_R		37					See fig. 10	See fig. 10			See fig. 10			1,3,5,7,11	5	V_R	V	0.10	1.30	V	0.10	1.30	V
	I_R		38						4/				5.0	4.5	2,3,5	5	I_R	mA	-1.6	---	mA	-1.6	---	mA

See footnotes at end of table.

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TABLE VIII. Group A inspection for device types 01, 02, and 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test No.	Adapter pin numbers 14/ 1	2 Device 01,03	2 Device 02	3	4	5	6 Device 01,03	6 Device 02	7	8 Device 01,03	8 Device 02	Relays closed 14/	Pin measured No.	Pin measured Value	Pin measured Unit	Device 01, 03 limits Min	Device 01, 03 limits Max	Device 01, 03 limits Unit	Device 02 limits 1/ Min	Device 02 limits 1/ Max	Device 02 limits 1/ Unit
					Volts, dc								Volts, dc											
3 T_C = -55°C	I_{CC}	3005	39	GND	4.5	5							3.5	3.8	2, 10	2	I_{CC}	mA	---	5.0	mA	---	10.0	mA
		3005	40		16.5	15							11.5	10.5	2, 10	2	I_{CC}	mA	---	20.0	mA	---	30.0	mA
	V_{TR}		41		4.5	5		2/					2/	2/	2,3,5,10	11	V_{TR}	V	1.15	1.80	V	1.30	1.95	V
			42		16.5	15		2/					2/	2/			V_{TR}	V	5.00	5.80	V	4.50	5.30	V
	I_{TR}		43		16.5	15							5.0	4.5			I_{TR}	µA	-5.0	---	µA	-5.0	---	µA
	V_{TH}		44		4.5	5		3/					3/	3/			V_{TH}	V	2.60	3.40	V	2.95	3.75	V
			45		16.5	15		3/					3/	3/			V_{TH}	V	10.60	11.40	V	9.60	10.40	V
	I_{TH}		46		16.5	15							11.5	10.5			I_{TH}	µA	---	2.5	µA	---	2.5	µA
	V_{OL}	3007	47		4.5	5		5 mA					3.5	3.8		4 15/	V_O	V	---	0.350	V	---	0.350	V
			48		4.5	5		50 mA					3.5	3.8					---	2.60		---	2.60	
			49		16.5	15		10 mA					11.5	10.5					---	0.150		---	0.150	
			50					50 mA											---	0.500		---	0.500	
			51					100 mA											---	2.80		---	2.80	
	V_{OH}	3006	52		4.5	5		-100 mA 16/					1.0	1.15	3,5,10	4	V_O	V	2.20	---	V	2.70	---	V
		3006	53		16.5	15		-100 mA 16/					5.0	4.5	3,5,10	4	V_O	V	14.0	---	V	12.5	---	V
	I_{CEX}		54							16.5 V	15.0 V		5.0	4.5	2,10	6	I_O	nA	---	100	nA	---	100	nA
	V_{SAT}		55							50 mA	50 mA		11.5	10.5	2,10	6	V_{SAT}	V	---	0.80	V	---	0.80	V
	V_R		56					See fig. 10	See fig. 10			See fig. 10			1,3,5,7,11	5	V_R	V	0.10	1.30	V	0.10	1.30	V
	I_R		57						4/				5.0	4.5	2,3,5	5	I_R	mA	-1.60	---	mA	-1.60	---	mA
9 T_C = 25°C	t_{PLH}		58		4.5	5		See				See			1,3,5,7,10	4	t_{PLH}	ns	---	800	ns	---	800	ns
			59		16.5	15		fig. 7				fig. 7					t_{PLH}		---	800		---	800	
	t_{TLH}		60		4.5	5											t_{TLH}		---	300		---	300	
			61		16.5	15											t_{TLH}		---			---		
	t_{THL}		62		4.5	5											t_{THL}		---			---		
			63		16.5	15											t_{THL}		---			---		
	t_{PHL}		64		4.5	5		See				See			2,3,5,10,11		t_{PHL}	µs	---	12.0	µs	---	12.0	µs
			65		16.5	15		fig. 8				fig. 8			2,3,5,10,11		t_{PHL}		---	12.0		---	12.0	
	$t_{D(OH)}$		66		4.5	5		See				See			1,3,5,7,10,11		t_D (OH)		106.7	113.3		106.7	113.3	
			67		16.5	15		fig. 7				fig. 7			1,3,5,7,10,11				106.7	113.3		106.7	113.3	
			68		4.5	5									1,3,6,7,10,11			ms	10.67	11.33	ms	10.67	11.33	ms
			69		16.5	15									1,3,6,7,10,11			ms	10.67	11.33	ms	10.67	11.33	ms
	$\frac{\Delta t_{D(OH)}}{\Delta V_{CC}}$		70		ΔV_{CC}= 12	ΔV_{CC}= 10	For device 01 and 03, V_{CC} = 4.5 V to 16.5 V 5/(See fig.7) For device 02, V_{CC} = 5.0 V to 15.0 V 6/ (See fig.7)								1,3,5,7,10,11		$\frac{\Delta t_D (OH)}{\Delta V_{CC}}$	ns / V	-220	220	ns / V	-220	220	ns / V
	t_{ch}		71		4.5	5		See				See			1,2,3,5,8,10		t_{ch}	µs	120	156	µs	120	156	µs
			72		16.5	15		fig. 9				fig. 9			1,2,3,5,8,10			µs	120	156	µs	120	156	µs
			73		4.5	5									1,2,3,6,9,10			ms	11.3	15.0	ms	11.3	15.0	ms
			74		16.5	15									1,2,3,6,9,10			ms	11.3	15.0	ms	11.3	15.0	ms
	t_{dis}		75		4.5	5									1,2,3,5,8,10		t_{dis}	µs	57.5	80.0	µs	57.5	80.0	µs
			76		16.5	15									1,2,3,5,8,10			µs	57.5	80.0	µs	57.5	80.0	µs
			77		4.5	5									1,2,3,6,9,10			ms	5.40	7.70	ms	5.40	7.70	ms
			78		16.5	15									1,2,3,6,9,10			ms	5.40	7.70	ms	5.40	7.70	ms
	$\frac{\Delta t_{ch}}{\Delta V_{CC}}$		79		ΔV_{CC}= 12	ΔV_{CC}= 10	For device 01 and 03, V_{CC} = 4.5 V to 16.5 V 7/(See fig.9) For device 02, V_{CC} = 5.0 V to 15.0 V 8/ (See fig.9)										$\frac{\Delta t_{ch}}{\Delta V_{CC}}$	ns / V	-820	820	ns / V	-820	820	ns / V
	$\Delta t_{D(OH)M}$		80		---	5	For device 02 only 11/ (See fig. 7)								1,3,5,7,10,11		Δt_D (OH)M	µs	---	---	---	-1.1	+1.1	µs
			81		---	15	For device 02 only 11/ (See fig. 7)								1,3,5,7,10,11			µs	---	---	---	-1.1	+1.1	µs

See footnotes at end of table.

MIL-M-38510/104A

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TABLE III. Group A inspection for device types 01, 02, and 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test No.	Adapter pin numbers 14/ 1	2 Device 01,03 (Volts, dc)	2 Device 02 (Volts, dc)	3	4	5	6 Device 01,03	6 Device 02	7	8 Device 01,03 (Volts, dc)	8 Device 02 (Volts, dc)	Relays closed 14/	Pin measured No.	Value	Unit	Device 01, 03 limits Min	Max	Unit	Device 02 limits 1/ Min	Max	Unit
9 T_C = 25°C	$\frac{\Delta t_{D(OH)M}}{\Delta V_{CC}}$		82	GND	---	V_{CC}= 10	For device 02 only, V_{CC} = 5.0 V to 15.0 V 12/ (See fig. 7)								1,3,5,7,10,11	4	$\frac{\Delta t_{D(OH)M}}{\Delta V_{CC}}$	ns / V	---	---	---	-110	+110	ns / V
	t_{res}		83		16.5	15	See fig. 10					See fig. 10			1,3,5,7,11		t_{res}	µs	---	1.5	µs	---	1.5	µs
10 T_C = 125°C	t_{PLH}		84		4.5	5	See fig. 7					See fig. 7			1,3,5,7,10,11		t_{PLH}	ns	---	900	ns	---	900	ns
			85		16.5	15											t_{PLH}			900			900	
	t_{TLH}		86		4.5	5											t_{TLH}		---	300		---	300	
			87		16.5	15											t_{TLH}		---			---		
	t_{THL}		88		4.5	5											t_{THL}		---			---		
			89		16.5	15											t_{THL}		---			---		
	t_{PHL}		90		4.5	5	See fig. 8					See fig. 8			2,3,5,10,11		t_{PHL}	µs	---	12.0	µs	---	12.0	µs
			91		16.5	15	See fig. 8					See fig. 8			2,3,5,10,11		t_{PHL}		---	12.0		---	12.0	
	$t_{D(OH)}$		92		4.5	5	See fig. 7					See fig. 7			1,3,5,7,10,11		t_D (OH)		106.7	113.3		106.7	113.3	
			93		16.5	15									1,3,5,7,10,11				106.7	113.3		106.7	113.3	
			94		4.5	5									1,3,6,7,10,11			ms	10.67	11.33	ms	10.67	11.33	ms
			95		16.5	15									1,3,6,7,10,11			ms	10.67	11.33	ms	10.67	11.33	ms
	$\frac{\Delta t_{D(OH)}}{\Delta T}$		96		16.5	15							See fig. 7 9/		1,3,5,7,10,11		$\frac{\Delta t_D}{\Delta T}$ (OH)	ns / °C	-11	11	ns / °C	-11	11	ns / °C
	t_{ch}		97		4.5	5	See fig. 9					See fig. 9			1,2,3,5,8,10		t_{ch}	µs	120	156	µs	120	156	µs
			98		16.5	15									1,2,3,5,8,10			µs	120	156	µs	120	156	µs
			99		4.5	5									1,2,3,6,9,10			ms	11.3	15.0	ms	11.3	15.0	ms
			100		16.5	15									1,2,3,6,9,10			ms	11.3	15.0	ms	11.3	15.0	ms
	t_{dis}		101		4.5	5									1,2,3,5,8,10		t_{dis}	µs	57.5	80.0	µs	57.5	80.0	µs
			102		16.5	15									1,2,3,5,8,10			µs	57.5	80.0	µs	57.5	80.0	µs
			103		4.5	5									1,2,3,6,9,10			ms	5.40	7.70	ms	5.40	7.70	ms
			104		16.5	15									1,2,3,6,9,10			ms	5.40	7.70	ms	5.40	7.70	ms
	$\frac{\Delta t_{ch}}{\Delta T}$		105		16.5	15	10/ (See fig. 9)								1,2,3,5,8,10		$\frac{\Delta t_{ch}}{\Delta T}$	ns / °C	-68	68	ns / °C	-68	68	ns / °C
	$\frac{\Delta t_{D(OH)M}}{\Delta T}$		106		---	15	For device 02 only 13/ (See fig. 7)								1,3,5,7,10,11		$\frac{\Delta t_D}{\Delta T}$ (OH)M	ns / °C	---	---	---	-20	+20	ns / °C
	t_{res}		107		16.5	15		See fig. 10				See fig. 10			1,3,5,7,11		t_{res}	µs	---	2.0	µs	---	2.0	µs
11 T_C = -55°C	t_{PLH}		108		4.5	5	See fig. 7					See fig. 7			1,3,5,7,10,11		t_{PLH}	ns	---	800	ns	---	800	ns
			109		16.5	15											t_{PLH}		---	800		---	800	
	t_{TLH}		110		4.5	5											t_{TLH}		---	300		---	300	
			111		16.5	15											t_{TLH}		---			---		
	t_{THL}		112		4.5	5											t_{THL}		---			---		
			113		16.5	15											t_{THL}		---			---		
	t_{PHL}		114		4.5	5	See fig. 8					See fig. 8			2,3,5,10,11		t_{PHL}	µs	---	12.0	µs	---	12.0	µs
			115		16.5	15	See fig. 8					See fig. 8			2,3,5,10,11		t_{PHL}		---	12.0		---	12.0	
	$t_{D(OH)}$		116		4.5	5	See fig. 7					See fig. 7			1,3,5,7,10,11		t_D (OH)		106.7	113.3		106.7	113.3	
			117		16.5	15									1,3,5,7,10,11				106.7	113.3		106.7	113.3	
			118		4.5	5									1,3,6,7,10,11			ms	10.67	11.33	ms	10.67	11.33	ms
			119		16.5	15									1,3,6,7,10,11			ms	10.67	11.33	ms	10.67	11.33	ms

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01, 02, and 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test No.	Adapter pin numbers 14/ 1	2 Device 01,03	2 Device 02	3	4	5	6 Device 01,03	6 Device 02	7	8 Device 01,03	8 Device 02	Relays closed 14/	Pin measured No.	Pin measured Value	Pin measured Unit	Device 01, 03 limits Min	Device 01, 03 limits Max	Device 01, 03 limits Unit	Device 02 limits 1/ Min	Device 02 limits 1/ Max	Device 02 limits 1/ Unit
					Volts, dc								Volts, dc											
11 T_C = -55°C	$\frac{\Delta t_{D(OH)}}{\Delta T}$		120	GND	16.5	15		See fig. 7				See fig. 7 9/	See fig. 7		1,3,5,7,10, 11	4	$\frac{\Delta t_{D(OH)}}{\Delta T}$	ns/°C	-11	11	ns/°C	-11	11	ns/°C
	t_{ch}		121		4.5	5		See				See			1,2,3,5,8,10		t_{ch}	µs	120	156	µs	120	156	µs
			122		16.5	15		fig. 9				fig. 9			1,2,3,5,8,10			µs	120	156	µs	120	156	µs
			123		4.5	5									1,2,3,6,9,10			ms	11.3	15.0	ms	11.3	15.0	ms
			124		16.5	15									1,2,3,6,9,10			ms	11.3	15.0	ms	11.3	15.0	ms
	t_{dis}		125		4.5	5									1,2,3,5,8,10		t_{dis}	µs	57.5	80.0	µs	57.5	80.0	µs
			126		16.5	15									1,2,3,5,8,10			µs	57.5	80.0	µs	57.5	80.0	µs
			127		4.5	5									1,2,3,6,9,10			ms	5.40	7.70	ms	5.40	7.70	ms
			128		16.5	15									1,2,3,6,9,10			ms	5.40	7.70	ms	5.40	7.70	ms
	$\frac{\Delta t_{ch}}{\Delta T}$		129		16.5	15	10/ (See fig. 9)								1,2,3,5,8,10		$\frac{\Delta t_{ch}}{\Delta T}$	ns/°C	-68	68	ns/°C	-68	68	ns/°C
	$\frac{\Delta t_{D(OH)M}}{\Delta T}$		130		---	15	For device 02 only 13/ (See fig. 7)								1,3,5,7,10, 11		$\frac{\Delta t_{D(OH)M}}{\Delta T}$	ns/°C	---	---	---	-20.0	+20	ns/°C
	t_{res}		131		16.5	15			See fig. 10			See fig. 10			1,3,5,7,11		t_{res}	µs	---	1.5	µs	---	1.5	µs

1/ Limits apply to both halves of device type 02 independently.

2/ The voltage at adapter pin 8 is swept downward from a voltage > 2/3 V_{CC}. V_{TR} is measured when the output switches to the "high" state.

3/ The voltage at adapter pin 8 is swept upward from a voltage < 1/3 V_{CC}. V_{TH} is measured when the output switches to the "low" state.

4/ The application of GND to test circuit pin 5 causes the output to be reset (latched low). Measure I_R into adapter pin 5.

5/ $\frac{\Delta t_{D(OH)}}{\Delta V_{CC}} = \frac{t_{D(OH)} \text{ at } V_{CC} = 16.5\text{ V} - t_{D(OH)} \text{ at } V_{CC} = 4.5\text{ V}}{16.5\text{ V} - 4.5\text{ V}}$

6/ (For 02) $\frac{\Delta t_{D(OH)}}{\Delta V_{CC}} = \frac{t_{D(OH)} \text{ at } V_{CC} = 15\text{ V} - t_{D(OH)} \text{ at } V_{CC} = 5\text{ V}}{15\text{ V} - 5\text{ V}}$

7/ $\frac{\Delta t_{ch}}{\Delta V_{CC}} = \frac{t_{ch} \text{ at } V_{CC} = 16.5\text{ V} - t_{ch} \text{ at } 4.5\text{ V}}{16.5\text{ V} - 4.5\text{ V}}$

8/ (For 02) $\frac{\Delta t_{ch}}{\Delta V_{CC}} = \frac{t_{ch} \text{ at } V_{CC} = 15\text{ V} - t_{ch} \text{ at } 5\text{ V}}{15\text{ V} - 5\text{ V}}$

9/ $\frac{\Delta t_{D(OH)}}{\Delta T} = \frac{t_{D(OH)} \text{ at } 125°C - t_{D(OH)} \text{ at } 25°C}{125°C - 25°C}$ and $= \frac{t_{D(OH)} \text{ at } 25°C - t_{D(OH)} \text{ at } -55°C}{25°C - (-55°C)}$

10/ $\frac{\Delta t_{ch}}{\Delta T} = \frac{t_{ch} \text{ at } 125°C - t_{ch} \text{ at } 25°C}{125°C - 25°C}$ and $= \frac{t_{ch} \text{ at } 25°C - t_{ch} \text{ at } -55°C}{25°C - (-55°C)}$

11/ $\Delta t_{D(OH)M} = t_{D(OH)}$ side "A" - $t_{D(OH)}$ side "B"

12/ $\frac{\Delta t_{D(OH)M}}{\Delta V_{CC}} = \frac{\Delta t_{D(OH)}}{\Delta V_{CC}}$ side "A" - $\frac{\Delta t_{D(OH)}}{\Delta V_{CC}}$ side "B"

13/ $\frac{\Delta t_{D(OH)M}}{\Delta T} = \frac{\Delta t_{D(OH)}}{\Delta T}$ side "A" - $\frac{\Delta t_{D(OH)}}{\Delta T}$ side "B"

14/ Test circuit pin numbers and relays refer to those shown on figure 6.

15/ V_{OL} shall be measured after the output has been low for 100 µs (minimum).

16/ I_{SOURCE} for device type 03 = -60 mA.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.
b. Life tests for class B and C devices (method 1005 of MIL-STD-883) - Test condition C using the circuits shown on figure 4, or test condition F using the circuits shown on figure 5.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given are referenced to the external zero reference level of the supply voltage. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias.

TABLE IV. Group C end-point electrical parameters.
(T_C = 25°C, V_{CC} = 16.5 Vdc for device types 01 and 03, V_{CC} = 15.0 Vdc for device type 02)

Test	Limits for device types 01 and 03		Limits for device type 02		Delta limits 1/
	Min	Max	Min	Max	
V_{TR}	5.20 V	5.80 V	4.70 V	5.30 V	±50 mV
V_{TH}	10.70 V	11.30 V	9.70 V	10.30 V	±50 mV
V_{OL} at 10 mA	-	0.150 V	-	0.150 V	±50 mV
I_{CEX}	-	100 nA	-	100 nA	±50 nA

1/ Delta limits apply to the measured value (see delta limit definition in MIL-M-38510).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).
b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirements for certificate of compliance, if applicable.

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d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
e. Requirements for packaging and packing.
f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.
g. Requirements for product assurance options.
h. Requirements for carriers, special lead lengths or lead forming. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

V_{TR} - - - - - - - - - Trigger voltage. The voltage at which the output latches from the "low" state to the "high" state. This voltage is nominally 1/3 V_{CC}.

I_{TR} - - - - - - - - - Trigger current. The current flowing out of the trigger terminal while the output is in the "high" state.

V_{TH} - - - - - - - - - Threshold voltage. The voltage at which the output latches from the "high" state to the "low" state. This voltage is nominally 2/3 V_{CC}.

I_{TH} - - - - - - - - - Threshold current. The current flowing into the threshold terminal while the output is in the "low" state.

V_{CL} - - - - - - - - - Control voltage. The control voltage is the reference voltage for the threshold comparator. It is internally generated by a voltage divider (from V_{CC} to ground) tapped at 2/3 V_{CC}. NOTE: The divider is also tapped at 1/3 V_{CC}, which is the reference voltage for the trigger comparator.

V_R - - - - - - - - - Reset voltage. The reset acts as an inhibit. If $1.3 \leq V_R \leq V_{CC}$, the device is free to function. If $0\ V \leq V_R \leq 0.1\ V$, the output is forced to the "low" state. The output will remain low after the reset voltage goes high only if the trigger voltage is high. If the trigger voltage is low, the output will go high when the reset voltage goes high. The reset voltage going low also causes the discharge transistor to turn on, thus preventing the timing capacitor (C_T) from charging.

I_R - - - - - - - - - Reset current. The current out of the reset terminal after the reset voltage has been applied and the output is latched low.

V_{SAT} - - - - - - - - Discharge transistor saturation voltage. When the output is low, the discharge terminal is sinking current. V_{SAT} is defined as the collector-emitter voltage of the discharge transistor when sinking the specified current.

$t_{D(OH)}$ - - - - - - - Time delay, output high. In the monostable mode of operation, the interval of time the output remains high once triggered. This delay is given by the following equation: $t_{D(OH)} = 1.1\ R_T\ C_T$ (see figure 8).

t_{ch} - - - - - - - - - Capacitor charge time. In the astable mode of operation, the time interval during which the external timing capacitor (C_T) is charging from 1/3 V_{CC} to 2/3 V_{CC}. This interval is ideally given by the equation: $t_{ch} = 0.693\ (R_{TA}+R_{TB})\ C_T$ (see figure 9).

t_{dis} - - - - - - - - Capacitor discharge time. In the astable mode of operation, the time interval during which the external timing capacitor (C_T) is discharging from 2/3 V_{CC} to 1/3 V_{CC}. This interval is ideally given by the equation: $t_{dis} = 0.693\ R_{TB}\ C_T$ (see figure 9).

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6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	555
02	556
03	555 (I_{SOURCE} = -60 mA)

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NS

Review activities:
Army - MI, AR
Air Force - 11, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-0377)

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	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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